

Stress-Testing Product-Gated Clustering and Bounded Priority Ranking for Flood-Rescue Reports

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Abstract. Operational flood streams contain duplicate, incomplete, and adversarial reports, yet clustering and priority scores are often evaluated separately from scheduling consequences. We study a fail-closed batch pipeline: receipt-visible reports with location and event time enter a sparse similarity graph; unresolved reports enter review; deterministic duplicate families feed a bounded ranking; and evaluator-only truth joins after predicted-cluster scheduling. The current draft clustering benchmark uses 80 independently seeded synthetic datasets geographically anchored to the Copernicus EMSR848 activation: 20 development, 20 calibration, and 40 held-out test runs from one fixed generator regime. Additive Louvain has mean ARI 0.9191, product Leiden 0.9165, and product Louvain 0.9072. The paired product-minus-additive difference is -0.0118 with bootstrap 95% CI [-0.0226,-0.0024], while the Holm-adjusted Wilcoxon result does not establish a difference ($p = 0.1975$). Ranking and scheduling remain part of the proposed pipeline, but their numerical results are pending dedicated benchmarks. Neither mathematical boundedness nor this synthetic clustering result implies policy validity, field benefit, OOD transfer, or misinformation robustness.

Keywords: Flood rescue · Graph clustering · Synthetic stress test · Priority ranking · Negative results.

1 Introduction and Related Work

Crisis streams mix repeated observations of one event, nearby but distinct events, missing fields, and nonincident noise. Social-sensor and emergency message research has long emphasized volume, verification, and event extraction [14,3,12]. Multimodal collections such as CrisisMMD and FloodNet capture important disaster modalities [1,13], but they do not jointly provide fine-grained physical-incident linkage, rescue priority, and dispatch outcomes. Consequently, high classification accuracy cannot by itself establish that a report becomes the right operational destination.

Graph community detection [5,16] and density methods such as DBSCAN, HDBSCAN, and ST-DBSCAN [8,6,4] offer plausible consolidation mechanisms. Multiplicative range/domain terms also occur in bilateral filtering [15]; we therefore do not claim a new generic kernel. Spatially constrained clustering and regionalization encode geography through different objectives [7,2]. Humanitarian

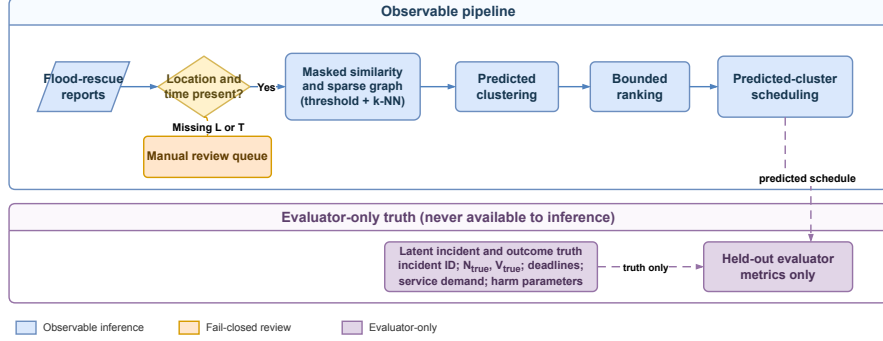


Fig. 1. Observable pipeline (solid), manual-review path (orange), and evaluator-only truth join (dashed). The scheduler receives predicted clusters, never incident truth.

scheduling further exposes conflicting equity, efficacy, and efficiency objectives [9,10,17]. A bounded score is thus a numerical property, not an expert-endorsed rescue policy.

We ask three questions: (RQ1) does independently selected product composition improve clustering over an additive composition under disclosed search spaces and one selection rule? (RQ2) does a bounded, duplicate-aware score align with an independently generated benefit and resist missingness, duplicates, contradictions, and campaigns? (RQ3) what happens when predicted split, merge, and noise errors propagate into dispatch rather than being repaired by incident truth? The present draft populates RQ1 from the available benchmark; RQ2 and RQ3 remain registered targets pending dedicated experimental outputs.

The contributions are: (i) an observable-only pipeline with explicit review and evaluator boundaries; (ii) complete product and additive threshold-domain statements, a component corollary conditional on observed hop diameter, and a duplicate-aware bounded ranking; and (iii) a development–calibration–held-out benchmark over 80 synthetic datasets from one fixed generator regime. The main contribution is diagnostic: mathematical safeguards and held-out synthetic accuracy must not be converted into deployment claims.

2 Observable Pipeline and Methods

Figure 1 fixes the information boundary. Receipt-visible fields flow through graph construction, ranking, and a batch scheduler. Missing location or event time routes a report to review. Incident identity, benefit, deadline, and harm are stored separately and join only after a schedule exists.

2.1 Missingness-Aware Similarity and Graph

A report contains location L_i , event time T_i , flood evidence F_i , urgency evidence E_i , reported demand N_i , vulnerability V_i , provenance features, receipt time, and an observation mask. The graph admission indicator $a_i = \mathbf{1}\{L_i, T_i \text{ are observed}\}$. Thus $a_i a_j = 0$ prevents an edge involving a report whose geographic or temporal separation is undefined; such a report follows the manual-review path. Missing F_i or E_i is not zero-imputed: let $I_F(i, j), I_E(i, j)$ indicate that both reports share the corresponding observation and let $m_{ij} = I_F(i, j) + I_E(i, j)$; then

$$C_{ij} = \frac{m_{ij}}{2} \exp \left[-I_F(i, j) \frac{|F_i - F_j|}{\tau_F} - I_E(i, j) \frac{|E_i - E_j|}{\tau_E} \right], \quad C_{ij} = 0 \text{ if } m_{ij} = 0.$$

When $m_{ij} = 2$, both contextual fields are used; when $m_{ij} = 1$, the maximum contextual similarity is discounted to $1/2$; and when $m_{ij} = 0$, no contextual agreement is inferred. In the current 80-run artifact, every report has location, event time, flood, and urgency. Hence $a_i = 1$ and $m_{ij} = 2$ throughout the evaluated data, reducing the expression to

$$C_{ij} = \exp \left[-\frac{|F_i - F_j|}{\tau_F} - \frac{|E_i - E_j|}{\tau_E} \right].$$

The missingness extension is specified and unit-tested, but missingness robustness is not claimed as an empirical result of this benchmark. For Haversine distance d_{ij} and time difference Δt_{ij} , define

$$\begin{aligned} G_{ij} &= \exp[-d_{ij}^2/(2\sigma^2)], & T_{ij} &= \exp[-\Delta t_{ij}/\tau_t], \\ w_{ij}^\times &= a_i a_j G_{ij}(\beta T_{ij} + \gamma C_{ij}), & w_{ij}^+ &= a_i a_j (\alpha G_{ij} + \beta T_{ij} + \gamma C_{ij}). \end{aligned} \quad (1)$$

All scales $\sigma, \tau_t, \tau_F, \tau_E$ are strictly positive and the weights are nonnegative. The implementation first forms a common spatial candidate pool of at most $\max(64, 4k)$ neighbours per eligible report, including all boundary-distance ties before canonical truncation. It computes an empirical nonzero-weight quantile, retains strict above-threshold edges, keeps the union of endpoint-wise top- k edges, and applies Louvain. Product and additive candidates have the same pool and grid, but each family is selected independently; the estimand is therefore a pipeline comparison, not an operator-only causal effect.

Theorem 1 (Product edge localization). *Let $B = \beta + \gamma > 0$. If a product edge with threshold $0 < \theta < B$ is retained, then $d_{ij} < \sigma \sqrt{2 \log(B/\theta)}$. For $\theta \geq B$ the strict retained set is empty; for $\theta \leq 0$ the threshold alone gives no finite cutoff.*

Proof. Since $a_i a_j \leq 1$ and $T_{ij}, C_{ij} \leq 1$, $w_{ij}^\times \leq B \exp[-d_{ij}^2/(2\sigma^2)]$. Combining $\theta < w_{ij}^\times$ with positivity and taking logarithms gives the bound; the endpoint cases follow from the same inequality.

This is an edge statement. A component-diameter claim additionally needs a bounded observed hop diameter; long transitive chains remain possible.

Corollary 1 (Conditional component bound). *In the finite product region, a connected component with unweighted observed hop diameter $h < \infty$ and geographic diameter D satisfies $D < hr_\theta^\times$, where $r_\theta^\times = \sigma\sqrt{2\log(B/\theta)}$. A singleton has $h = D = 0$.*

The result follows from the triangle inequality along a path of at most h retained edges. It is a conditional post-hoc component bound, not an ex-ante compactness guarantee: the pipeline does not control h .

For the additive form, assume $\alpha > 0$. If $B < \theta < B + \alpha$, every retained edge satisfies

$$d_{ij} < r_\theta^+ = \sigma\sqrt{2\log\left(\frac{\alpha}{\theta - B}\right)}.$$

If $\theta \geq B + \alpha$, the strict retained set is empty; if $\theta \leq B$, no non-trivial threshold-induced geographic cutoff exists in general. Indeed, retention gives $\theta < \alpha G_{ij} + \beta T_{ij} + \gamma C_{ij} \leq \alpha G_{ij} + B$. Thus product composition has a finite geographic edge bound throughout its nonempty positive-threshold domain, whereas additive composition has such a bound only above the maximum non-geographic contribution. Neither statement alone implies better empirical clustering.

2.2 Duplicate-Aware Bounded Ranking

Provenance $Q_i \in [0, 1]$ combines a source prior, image presence, and capped corroboration from distinct source families; proximity alone cannot increase it. Exact duplicates share a canonical fingerprint. Near duplicates are merged by deterministic complete linkage, so $A \sim B \sim C$ does not merge A with C unless every cross-pair passes the envelope.

Let g index the resulting evidence families. Within each family, observed fields are provenance-gated; across families, demand and vulnerability use maxima rather than unsupported population sums. In compact form,

$$\begin{aligned} \bar{E} &= \frac{\sum_g \max_{i \in g} Q_i E_i}{\max(1, \sum_g \max_{i \in g} Q_i)}, & \bar{F} &= \max_{g, i \in g} Q_i F_i, \\ \bar{N} &= \min\left(1, \frac{\log(1 + \max_{g, i \in g} Q_i N_i)}{\log(1 + N_{\text{ref}})}\right), & \bar{V} &= \max_{g, i \in g} Q_i V_i, \\ P &= [1 + (\mu - 1) \tanh(\bar{V}/s)] (\omega_E \bar{E} + \omega_F \bar{F} + \omega_N \bar{N}). \end{aligned} \quad (2)$$

Claims are clipped before aggregation. We fix $\omega_E = .40$, $\omega_F = .35$, $\omega_N = .25$, and $\mu = 1.75$; hence $0 \leq P \leq 1.75$. The weights and caps are policy knobs, not learned or expert-validated parameters. Comparators are a multiplicity-sensitive legacy score, urgency-only, population-only, a simple linear score, stable random, and—for dispatch—nearest-first.

3 Experimental Design

Data contract and splits. The artifact contains 80 independently seeded synthetic datasets geographically anchored to the Copernicus EMSR848 Central

Viet Nam flood activation. Copernicus EMS, OpenStreetMap, and WorldPop provide geographic or historical anchors; rescue incidents, reports, duplicates, coordinated campaigns, vulnerability, and incident labels are synthetic. Every run has 16 latent incidents, 288–370 reports, exact and near duplicates, and three coordinated nonincident campaigns of 20 reports each. Across all runs are 26,288 reports: 17,901 genuine, 3,587 duplicate, and 4,800 coordinated nonincident reports. Runs 1–20 are reserved for development, runs 21–40 for calibration, and runs 41–80 for held-out testing. All runs use one fixed generator regime, so the experiment measures within-generator variation, not mechanism-shift OOD transfer. Inference reads only `algorithm_input.json`; ground truth is opened after predictions for a test run have been produced.

Selection and comparators. The current notebook evaluates complete small grids on the 20 calibration runs and selects mean ARI, breaking ties by standard deviation and canonical configuration JSON. Product and additive use fixed $\tau_F = .25$, $\tau_E = .35$, and $\beta = \gamma = .5$. Selected product Louvain uses $\sigma = 700$ m, $\tau_t = 60$ min, quantile .90, $k = 8$, and resolution 1.2. Selected additive Louvain uses $\alpha = 1$, $\sigma = 700$ m, $\tau_t = 60$ min, quantile .95, $k = 8$, and resolution 1.2. Because selected quantiles differ, this is an independently calibrated pipeline comparison, not a matched-density operator comparison. Comparators are a convex similarity, product Leiden, geo-time DBSCAN, DBSCAN/HDBSCAN on scaled observable features, spatially constrained agglomerative clustering, and coordinate K-means. The implemented geo-time DBSCAN lacks ST-DBSCAN’s non-spatial criterion and is named accordingly.

Endpoints and inference. ARI on incident-linked reports [11] is the current primary draft endpoint; NMI, pairwise F1, geographic diameter, and runtime are descriptive. The existing noise-absorption implementation omits fake-only clusters from false destinations, so that metric is not reported as evidence. Before freezing the benchmark, the implementation must add false destinations, fake absorption, noise rejection, retained-edge fraction, mean degree, and a matched-density product–additive comparison. Seed-paired ARI differences use 10,000 bootstrap resamples, Wilcoxon signed-rank tests, and Holm correction. The registered priority stress tests and predicted-cluster dispatch evaluation remain unchanged in scope, but their numerical results are pending independent benefit, deadline, service, access-delay, and harm outputs. No clustering metric is substituted for those outcomes.

4 Results

Table 1 gives the current replaceable RQ1 draft. Additive Louvain has the highest mean ARI, followed by product Leiden, the convex composition, and product Louvain. Product Louvain minus additive Louvain has mean paired ARI difference -0.01184 with bootstrap 95% CI $[-0.02257, -0.00240]$: product records 9 wins, 19 ties, and 12 losses. The raw Wilcoxon p is 0.0987 and its Holm-adjusted

Table 1. Draft held-out clustering snapshot over runs 41–80. ARI uncertainty is the reported normal-approximation half-width. Lower diameter is descriptive, not a co-primary endpoint. These values will be replaced after the benchmark adds matched-density and operational-noise metrics.

Method	ARI 95% half-width	Pair F1	Mean diam. (km)
Additive Louvain	0.9191	0.0197 0.9247	0.3415
Product Leiden	0.9165	0.0190 0.9223	0.3547
Convex Louvain	0.9135	0.0192 0.9195	0.3547
Product Louvain	0.9072	0.0205 0.9138	0.3585
Coordinate K-means	0.8176	0.0203 0.8292	0.2972
Spatial agglomerative	0.6235	0.0269 0.6506	4.1648
HDBSCAN	0.5205	0.0628 0.5632	1.2527
Geo-time DBSCAN	0.4978	0.0390 0.5450	0.6244
DBSCAN, all features	0.4741	0.0471 0.5196	0.9340

value is 0.1975. Thus the registered multiplicity-adjusted test does not establish a product–additive difference. Product Leiden also does not differ from product Louvain after Holm adjustment ($p = 0.0840$).

The graph compositions have substantially higher held-out ARI than the current density and agglomerative implementations, but the result is limited to their disclosed grids and the single authored generator regime. RQ2 priority and RQ3 dispatch tables are intentionally pending. They must not be filled from the earlier estimated ID/OOD values or inferred from clustering ARI. The final version will insert results only after dedicated benchmark artifacts provide independent ranking benefit and scheduling outcomes.

5 Discussion and Threats to Validity

The experiment separates mathematical safeguards from empirical benefit. Product gating localizes retained edges, yet its independently selected pipeline has a lower mean ARI than additive in the present draft, without a Holm-significant paired difference. Because product and additive selected different weight quantiles, their contrast is not an isolated causal effect of multiplication. A matched-density comparison is required. Likewise, score boundedness and duplicate-aware aggregation limit numerical influence but do not by themselves establish source authenticity or priority validity; those properties still require the registered RQ2 stress tests.

All reports and incident linkage are synthetic. The 40 test runs vary random seeds inside one generator regime rather than changing mechanisms. The benchmark therefore provides neither OOD nor field validation. All evaluated reports contain location, event time, flood, and urgency, so the missingness extension has not been empirically tested. The current noise metric fails to count fake-only clusters as false destinations and is omitted until corrected. No expert panel val-

idated priority weights, caps, or operational utility. No current result supports “deployment-ready”, misinformation robustness, or real-world harm reduction.

6 Artifact and Conclusion

The accompanying draft artifact contains the 80-run dataset, benchmark notebook, calibration grid, selected configurations, per-run test results, paired comparisons, and editable Figure 1. The current table is a replaceable snapshot. Before submission, every number must be generated from the accepted CSV and bound to code, data, and configuration hashes.

In conclusion, the complete threshold analysis identifies the finite geographic domains of product and additive graph weights and makes component compactness conditional on observed hop diameter. In the current held-out draft, additive Louvain has the highest mean ARI, but multiplicity-adjusted paired inference does not establish a difference from product Louvain. The full observable pipeline, bounded priority ranking, and predicted-cluster scheduling remain part of the proposed study; their RQ2/RQ3 results await dedicated benchmark artifacts. The defensible current result is competitive within-generator clustering plus mathematical audit properties, not policy or deployment validation.

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